

Harry Nguyen

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EDUCATION

Georgia State University

Expected May 2027

Bachelor of Science in Computer Science • **GPA: 3.75/4.0**

- Coursework: Distributed Systems, Operating Systems, Database Systems, Computer Networks, Data Structures
- Honors: Presidential Scholarship, Anita Andrew Fellowship, Roxie Alexander Memorial Award
- Activities: Amazon University Event – BEN Tech Cloud Connect ATL

EXPERIENCE

TiMoto AI | Software Engineer

Sep 2025 – Present

- Built **LLM-as-a-judge evaluation pipeline** – AI-based automated assessment before deployment; **LangChain** orchestration + Claude/GPT LLM API integration; **sub-50ms p99** inference SLO; automated regression detection replacing manual review at scale
- Designed **gRPC inter-service layer** with exactly-once delivery and circuit breaker auto-rollback – traced production deadlock to shared resource acquisition conflicts; redesigned call sequencing; assessed failure modes of distributed systems under concurrent load; **100% evaluation success rate**
- Deployed **vLLM with PagedAttention** for production **ML infrastructure** – solved KV cache memory fragmentation under concurrent load; **zero OOM failures**; continuous batching throughput optimization; benchmarked vs. naive HF inference on throughput, p99, and memory utilization
- Analyzed production telemetry (**CloudWatch + Prometheus/Grafana**) – identified bottlenecks, implemented durable fixes; triaged incidents via root-cause analysis, runbooks, post-mortem; **99.9% uptime SLO**; **44% infra cost reduction** (\$40–60/mo)

Google | Software Engineer Intern – Chrome Browser

May 2025 – Aug 2025

- Designed **C++ IPC transport layer** with **Protocol Buffers** at Google's security review standard – evaluated Protobuf vs. custom serialization (schema evolution, memory safety, cross-language compatibility); shipped to Chrome stable serving **3B+ active users** at **sub-50ms p99**; zero production regressions at **95% test coverage**
- Diagnosed p99 bottleneck at 1,200ms; profiled critical path; implemented **lock-free concurrent trie search** with **CAS operations** eliminating mutex contention – **96% latency reduction**, zero production regressions; changes adopted by senior Chrome engineers
- Architected event-driven **TypeScript/React** system with observer pattern – **68% feature delivery acceleration** across **25K+ lines** of Chromium at **95% test coverage**; design documents and code reviews adopted into production branch

Develop for Good | Software Engineer Intern

May 2024 – Aug 2024

- Designed stateless **secure BaaS on AWS** – JWT over session auth for horizontal scalability and secure stateless design; auto-scaling supporting **500+ concurrent users**; **90% deployment time reduction** via CI/CD (GitHub Actions)
- Diagnosed N+1 query bottleneck degrading response to 3s+; redesigned with **PostgreSQL composite indexing** achieving **sub-100ms** for 10,000+ records

PROJECTS

Pulumi – Contributor | *Go, TypeScript, Infrastructure-as-Code*

github.com/pulumi

- Submitted **Go CLI features** for multi-cloud (AWS/Azure/GCP) provisioning – open-source developer tooling under active review by core maintainers
- Analyzed **Raft/Paxos consensus** in distributed state synchronization – studied linearizability guarantees and correctness under concurrent operations and partial failures

SKILLS

ML & AI Infrastructure: vLLM, PagedAttention, Continuous Batching, LLM-as-a-judge evaluation, LangChain, LLM APIs (Claude, GPT), PyTorch, LoRA/QLoRA

Distributed Systems: gRPC, Protocol Buffers, Exactly-once semantics, Circuit breakers, Multi-AZ, Fault tolerance, Event-driven architecture

Languages: Python, C++, Go, TypeScript, Java, JavaScript, SQL, Bash, Rust

Cloud & Infrastructure: AWS (ECS Fargate, EKS, EC2, S3, RDS), Terraform, Kubernetes, Docker, CI/CD (GitHub Actions), CloudWatch, Prometheus, Grafana

Frameworks & Databases: FastAPI, Django, Node.js, React, REST APIs, PostgreSQL, Redis, MongoDB

AI Dev Tools: Claude Code, GitHub Copilot, Codex, Cursor